

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	We exported dataset X from the UC Health COVID-19 Research Data Set (CORDS) database, exported dataset Y from the UCSD Health database, and downloaded the Edinburgh myocardial infarction data from the GitHub repository of ExplorerChain (http://doi.org/10.5281/zenodo.1492820). We adopted NIST JAMA (v1.0.3), Apache Common Math (v1.2), Apache Commons CSV (v1.9.0), and Java Utilities (v1.0.0) libraries. The data preprocessor to preprocess/split the raw data for Distributed Cross-Learning for Equitable Federated models (D-CLEF) is implemented in Java and available at https://doi.org/10.5281/zenodo.14052533 .
Data analysis	The D-CLEF software was implemented in Java using the following libraries: Go Ethereum (v1.9.1), Kubo IPFS (v0.20.0), NIST JAMA (v1.0.3), Apache Common Math (v1.2), jBLAS (v1.2.6 snapshot), Java Utilities (v1.0.0), and Web3j CLI (v1.4.1). The evaluation used AUC Calculator (v0.2) library. The software is available at https://doi.org/10.5281/zenodo.14052494 .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The University of California (UC) Health COVID-19 Research Data Set (CORDS, dataset X) are available under restricted access for containing patients' data, access can be obtained by UC affiliated researchers with permission of UC Biomedical Research Acceleration, Integration, & Development (UC BRAID, <https://www.ucbraid.org>) and Center for Data-Driven Insights and Innovation (CDI2, <https://ucop.edu/uc-health/index.html>) with upon approved Data Usage Agreement (DUA). The timeframe of response to requests and availability of data will be determined by UC BRAID and CDI2. The UC San Diego (UCSD) Total Hip Arthroplasty (THA) surgery data (dataset Y) are available under restricted access for containing patients' data, access can be obtained by UCSD affiliated researchers with permission of UCSD (<https://www.ucsd.edu>) upon approved Institutional Review Board (IRB). The timeframe of response to requests and availability of data will be determined by UCSD. The Edinburgh myocardial infarction data (dataset Z) used in this study are available in the Zenodo database under accession code: 10.5281/zenodo.1492820 (<http://doi.org/10.5281/zenodo.1492820>). Source data are provided with this paper.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

The gender information of dataset X was collected from the UC CORDS data and summarized in Supplementary Fig. 4b (8,032 females, 7,243 males, and 4 unknown ones). The sex information of dataset Y was collected from UCSD and aggregated in Supplementary Fig. 4e (477 male and 483 non-male). There is no sex/gender information collected for dataset Z. The findings of this study apply to all sex and gender. Sex/gender were determined based on self-reporting and were used as covariates (for dataset X and Y) in our study design and analysis. The University of California San Diego (UCSD) Human Research Protections Program (HRPP) Institutional Review Board (IRB) determined that this study is exempt from IRB requirements under category 45 CFR 46.104(d) (4) Secondary research, and consent was not required.

Reporting on race, ethnicity, or other socially relevant groupings

The race and ethnicity variables were included in dataset X as part of the data, and their statistics were summarized in Supplementary Fig. 4c and 4d, respectively. The race included eight values (American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White, Other Race, Multi Race, and Unknown). The ethnicity included three values (Hispanic or Latino, Not Hispanic or Latino, and Unknown). The English speaker variable was included in dataset Y as part of the data, and its statistics was summarized in Supplementary Fig. 4e. There were no race, ethnicity, or other socially relevant variables collected for dataset Z. For datasets X and Y, these variables were provided by the patients' self-reporting; were not used as proxies for other socially constructed/relevant variables such as socioeconomic status; and were part of the multivariate analyses to control confounding variables.

Population characteristics

The age group information was included in dataset X, and its statistics were summarized in Supplementary Fig. 4a with nine bins (0-10, 11-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80, and 81+). The elder (defined as whether age ≥ 65 years) variable was included in dataset Y, and its statistics were summarized in Supplementary Fig. 4e. There were no population characteristics variables collected for dataset Z.

Recruitment

We collected all data based on availability and there was no additional recruitment of participants. The potential risk of self-selection bias that could impact our results is relatively very low.

Ethics oversight

The use of all datasets was approved by UCSD HRPP IRB Exemption Category 4 (# 804237) on May 16, 2022.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

No statistical method was used to predetermine the sample size of each dataset. The sample sizes were chosen based on the available samples in each dataset (15,279 for dataset X, 960 for dataset Y, and 1,253 for dataset Z), which are sufficient for predictive modeling purposes.

Data exclusions

For dataset X, the non-pre-established exclusion criteria for data completeness include missing age, drug, breakthrough, or death information; unspecified, mixed, or other types of vaccine manufacturer; and unclear vaccination status. No data in dataset Y and X were excluded from

the analysis.

Replication

We randomly permuted and split each of our datasets into training and test sets for $n = 30$ trials independently to verify the reproducibility of our experimental findings, and all attempts at replication were successful.

Randomization

The samples were allocated into experimental groups based on patients' disease; no other covariates were used for allocation.

Blinding

The data were collected from pre-existing datasets and therefore blinding is not applicable.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.